

Doc. 262 – Acceptance without visualisation and inside-view of decompactified time

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Contents

Doc. 262 – Acceptance and inside-view	2
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Part A – Acceptance without visualisation **4**

.1	What is not visualisable is not unintelligible	4
.2	The actual acceptance criterion	5
.3	T^4 as a structural outside perspective	5
.4	What this is not	5
.5	Consequence for the discussion of FFGFT	6
.6	Summary	6

Part B – Time, mass, fractality **9**

.7	Time as mode, not stage	9
.8	Borrowed time: $\tilde{T}$ and energy as reciprocal readings	9
.9	Mass as deformation of the carrier	10
.10	T^4 as a scale-structural, not scale-metric, statement	10
.11	Fractality as an inside-view phenomenon	11
.12	Inside-view metaphor: brain folds and light	12
.13	Consequences	13
.14	Summary	13

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Document preface

This document brings together, in two complementary parts, the epistemic self-positioning of FFGFT regarding the visualisability of T^4 . **Part A** clarifies the negative side: spatial visualisability has, for over a century, not been an acceptance criterion of physics, and FFGFT stands in this tradition rather than outside it. **Part B** adds the positive side: what a deliberately delimited inside-view metaphor can deliver – up to the explicit framing of the brain-folds metaphor as an “inside-view metaphor, not an ontological claim”. The two parts belong together: without Part A, Part B would be misread as esotericism; without Part B, Part A would remain purely defensive.

**Part A – Acceptance without
visualisation: On the epistemic status
of T^4**

Preliminary note to Part A

A theory that posits a four-dimensional identification torus T^4 as the carrier of physics regularly encounters the objection: "I cannot visualise that." This part addresses the objection directly. It does not claim to deliver a spatial visualisation of T^4 – that is not possible. It shows instead that *visualisability* in the sense of spatial intuition has, for over a century, *not* been an acceptance criterion of physics, and that FFGFT stands within this methodological tradition, not outside it.

.1 What is not visualisable is not unintelligible

Modern physics routinely works with mathematical constructs that are not spatially visualisable, and accepts them because they work. A few examples:

- **Imaginary numbers.** $\sqrt{-1}$ was rejected for centuries as "impossible" – Descartes coined the derisive name "imaginary", which has stuck to this day. Today electrical engineering, signal processing and quantum mechanics use them as a matter of course. The decisive step towards acceptance was not spatial visualisation but Gauss's idea of representing them as a *direction perpendicular* to the real axis – a geometric extension that made the numbers tractable without "picturing" them in any everyday sense.
- **Complex-valued wave functions.** The wave function ψ of quantum mechanics is complex-valued; its imaginary component is physically effective (interference, phase, entanglement). No one can spatially visualise ψ , and yet it is the central object of all of quantum theory.
- **Minkowski spacetime.** In 1908 Minkowski introduced time with a factor of *ict* – as an *imaginary* coordinate – to handle Lorentz transformations cleanly in geometric terms. Standard relativity therefore treats time formally not like a spatial dimension; all of contemporary high-energy and astrophysics builds on this construction.
- **Infinite-dimensional Hilbert spaces.** Any serious formulation of quantum mechanics lives in an infinite-dimensional, typically complex Hilbert space. Spatial visualisability? Out of the question. Acceptance? Complete.
- **Gauge groups, spinors, Lie algebras.** The Standard Model of particle physics formulates itself over $SU(3) \times SU(2) \times U(1)$. Spinors are not vectors in the usual sense – they change sign under a 2π rotation, which seems intuitively absurd. Yet spinor mathematics is indispensable and its consequences are measured.
- **Complex manifolds, Calabi-Yau spaces.** String theory and parts of mathematical physics work with six- or ten-dimensional spaces whose geometry is not pictorial. They are accepted in their respective communities without anyone "seeing" them.
- **Negative numbers.** Rejected in medieval Europe because no one could picture "–3 apples". An apple is there or it is not; less than no apple is plainly nonsense. *Three apples borrowed*, however, was obvious even then – everyone understood what a debt is. The lesson of this story is not that negative numbers became more pictorial, but that the description itself had to change: not "picture a –3-apple" but *describe a debt relation structurally*. As soon as the descriptive frame shifted, the matter was immediately accepted.

The list is not exhaustive; it is not even representative. But it suffices to make the point: *visualisability in the sense of spatial intuition has, since the mathematisation of physics, not been a necessary acceptance criterion.*

.2 The actual acceptance criterion

If not spatial visualisability, then what? In the practice of physics since roughly Gauss and Minkowski, two criteria are established:

1. **Internal consistency.** The mathematical structures must be contradiction-free. Theorems must follow from definitions and axioms; derived quantities must behave in well-defined ways.
2. **Testable consequences.** The construction must deliver statements that can be compared with observation or measurement – and stand or fall in that comparison.

Both criteria are *structural*, not visual. They concern the mathematical and empirical behaviour of the theory, not its appearance before the reader's inner eye. A theory that satisfies both criteria is physically acceptable, regardless of whether its subject can be drawn.

This shift in the acceptance criterion is not mere convention. It is the methodological consequence of recognising that our visual system is evolutionarily tuned to a three-dimensional habitat and becomes systematically unreliable for structural claims beyond that habitat. Visualisability is a *psychological* phenomenon, not an epistemic one.

.3 T^4 as a structural outside perspective

What the constructs listed in §1 have in common is more than their unvisualisability. They are *structural outside perspectives* on systems that, from any single inside view, would be only partially visible.

- No observer sees *Lorentz geometry* – each sees only their own slice through it. But the Minkowski description gives us the geometry *as a whole*, within which the various observer perspectives appear as particular slices.
- No experimenter sees the *wave function* – each measurement yields only a single outcome. But the Hilbert-space description describes the wave function *as an object*, whose properties (norm, phase, superposition) stand the test of many measurements statistically.
- No one sees a *gauge group*. But the Standard-Model description captures the internal structure of the interactions *as group theory*, whose consequences (mass ratios, mixing angles, conservation laws) are measurable.

In each of these cases the lesson is the same: the mathematical *outside view* is richer than any possible inside experience. It makes visible what cannot be visible from within the system – and that is precisely its value.

T^4 **stands in this row.** It is not a spatial picture but a structural outside perspective on the carrier of physics in which quantum mechanics and relativity do not contradict each other. From any inside perspective – that is, from any concrete reference frame – QM and GR are incompatible; from the outside perspective T^4 they are different slices of the same closed structure. This is methodologically the same programme that Minkowski carried out for time, Hilbert for quantum mechanics, and the gauge theorists for internal symmetry: *not to produce an intuition, but to find an outside language.*

.4 What this is not

Three demarcations, so that the claim is not overstretched:

First: no privileged outside position. “Outside view” here means a different *descriptive perspective*, not a standpoint “outside physics”. We are never really outside; we change the descriptive frame. The mathematical outside view is not *the* outside view (no such thing exists) but *an* outside view – one that makes visible structures that would remain hidden from the inside.

Second: no ontological claim. FFGFT does not claim “the world is really a T^4 ”. It claims: *if we want to describe the physical system from an outside perspective in which the contradictory inside views of QM and GR can be unified, then T^4 is the minimal structure that achieves this.* This is a substantially weaker and at the same time more defensible claim than “this is how the world is”.

Third: no demand for visualisation. The aim of this document is not to produce an intuition of T^4 . That would be an empty promise. The aim is to separate the acceptance question from the visualisation question and to refer it back to the two criteria on which modern physics actually operates: consistency and testable consequences.

.5 Consequence for the discussion of FFGFT

Anyone who rejects FFGFT on the grounds that T^4 is not visualisable would, to be consistent, also have to reject:

- the complex-valued wave function (not visualisable),
- the Lorentz transformation with imaginary time (not visualisable),
- the infinite-dimensional Hilbert spaces (not visualisable),
- the spinors with 4π periodicity (not visualisable),
- the gauge groups of the Standard Model (not visualisable).

Since no one seriously demands this – and modern physics would collapse without these constructs – visualisability is not the decisive criterion in the case of T^4 either. The decisive criterion is whether FFGFT meets the two standard requirements:

1. **Internal consistency.** Developed in Doc. 001a (foundations), Doc. 181 (4D torus geometry), Doc. 193 (Non-Closure Theorem), Doc. 230 (Hilbert-space bridge).
2. **Testable consequences.** Concrete predictions for lepton masses and the CHSH deviation; the deviation from the Tsirelson bound lies below the measurement noise floor with present-day hardware (Doc. 147).

These two questions are the scientifically legitimate ones. The question of visualisability is not.

.6 Summary

Visualisability in the sense of spatial intuition has, for over a century, not been an acceptance criterion in physics. Internal consistency and testable consequences have taken its place. The mathematical constructs of modern physics – imaginary numbers, complex wave functions, Minkowski geometry, Hilbert spaces, gauge groups, spinors – are none of them spatially visualisable; they are nonetheless accepted because they supply structural outside perspectives on systems that, from any inside view, would remain incomplete.

T^4 stands in this row. It is not visualisable – and it need not be. It is precisely defined, internally consistent, and its consequences are derivable; some are already measured, others lie below the current measurement noise. That is the methodological situation. Anyone who rejects FFGFT for lack of visualisability rejects not FFGFT but the methodological programme of modern physics as a whole.

Imagination need not be visual to be effective.

**Part B – Time as mode, mass as
deformation, fractality as inside-view
of decompactified time**

Preliminary note to Part B

Part A established that T^4 is not spatially visualisable and need not be. That is the negative side: *what visualisation cannot deliver*.

This part makes the positive side explicit: *what an admitted inside-view metaphor can deliver*. It shows that the description of T^4 remains coherent when time is read not as a distinguished continuous variable but as a further mode on the torus – and that the fractal structure that appears in FFGFT is, in this reading, not a property of the underlying geometry but a phenomenon that arises in the transition to decompactified time. Within this transition – and only there – a particular inside-view metaphor becomes admissible, which is named explicitly and framed carefully at the end.

.7 Time as mode, not stage

On T^4 all four coordinates are equally periodic. None of them is distinguished as “time”; the division into “three spatial dimensions plus one temporal dimension” is not laid down in the carrier itself, but arises only from the choice of a descriptive perspective. In the FFGFT reading, what we phenomenally experience as *flowing time* is a *mode* in the sense of an oscillation or motion pattern on the torus – not a stage on which physics takes place.

This position is consistent with two points already established in the corpus:

- In FFGFT, $\hbar$ is an SI conversion factor between energy and time measures, not an ontological primitive (Doc. 001a, Doc. 230 § Operational equivalence). The status of time itself is therefore not primitive either, but derived.
- The foundational relation $\tilde{T} \cdot m = 1$ is, on T^4 , a strict reciprocity, not an open time coordinate. Reciprocity is a property of a closed structure, not of a flow.

Consequence. What we perceive as “continuously flowing time” is the phenomenal resolution of a discrete torus mode into a quasi-continuous variable. This resolution we call *decompactification of time*. It is not wrong – it delivers exactly the familiar predictions of standard physics – but it is a *descriptive choice*, not a property of the carrier.

.8 Borrowed time: $\tilde{T}$ and energy as reciprocal readings

Before treating the deformation of the carrier by mass, an observation is in order that clarifies the status of $\tilde{T}$ itself – and that leads into a very old didactic parallel.

The historical parallel. In medieval Europe, negative numbers were rejected because no one could picture “–3 apples”. Three apples borrowed, on the other hand, were obvious even then. Acceptance came not through a better picture but through a shift in the descriptive frame: not *inventory of apple objects* but *debt relation*. As soon as the frame shifted, the matter was immediately intelligible (cf. Part A § 1).

The transfer to $\tilde{T}$. What in the apple case is the shift from *object inventory* to *debt relation* is, in the case of time, the shift from *flowing time* to *borrowed energy*. On T^4 , $\tilde{T}$ is not an open time coordinate but a *reciprocal reading of the energetic action* on the carrier. The foundational relation

$$\tilde{T} \cdot m = 1 \tag{1}$$

says exactly this: $\tilde{T}$ and m are not two distinct ontological quantities but *two reciprocal readings of the same fundamental action*. A short $\tilde{T}$ corresponds to a high energy, a long $\tilde{T}$ to a low one – carrying the same information, in inverse scaling. This is not mathematically

new (in natural units, energy is simply reciprocal time), but in FFGFT it is taken ontologically seriously: $\tilde{T}$ is *borrowed energy*, not an independent “something”.

What follows from this. All dynamical quantities on T^4 can, in this reading, be understood as different scaling axes of a *single energetic action*: time ($\tilde{T}$), mass-energy (m, E), momentum, frequency, wave number. They differ not in their ontological category but in the choice of scaling axis used to express them. What remains as non-energetic substrate is only the *topology* of T^4 : the four cyclic identifications, the periodicity, the closure. This topology is not energy – it is what energy can be carried on.

The methodological point. As with borrowed apples, acceptance is a question of the descriptive frame, not of intuition. No one can spatially picture “borrowed time” – no more than “–3 apples”. But the structural notion of a reciprocal debt relation is, in both cases, clear and tractable. $\tilde{T} \cdot m = 1$ is, on T^4 , what the debt relation is in the apple case: a structural relation that needs no visualisation to function.

This reading prepares the following section: if $\tilde{T}$ and m are reciprocal readings of the same energetic action, then the “deformation of the carrier by mass” is nothing other than a local redistribution of this action over the geometry. Curvature and energy are not two different things that interact; they are two aspects of the same.

.9 Mass as deformation of the carrier

In general relativity, mass is described as curvature of the spacetime manifold. On T^4 , this idea carries over structurally: what GR reads as local curvature of an open manifold becomes, on the closed carrier, a local *deformation* of the torus geometry. The global topology is preserved – the four identifications continue to close the carrier – but the local metric is altered by mass.

Three remarks:

- In the limit of small deformation (local inertial frames), the deformed torus is indistinguishable from the Euclidean spacetime of GR – GR is, in this reading, the linear approximation in which the global closure is invisible.
- GR singularities (black hole, Big Bang) appear structurally differently on T^4 because the carrier is closed through identifications (cf. Doc. 230 § Operational equivalence, the point on singularities).
- The deformation is not visualisable – for exactly the reasons developed in Doc. 162. What is described here is the *structural relationship* between mass and carrier, not a picture.

.10 T^4 as a scale-structural, not scale-metric, statement

Up to this point T^4 has been treated as though it were *one* geometry of a fixed size. The consequences of the model are drawn, however, on very different length scales: lepton masses ($\sim 10^{-18}$ m), Bell correlations (laboratory, m to km), material dispersion in LiNbO_3 ($\sim \mu\text{m}$), Hubble scale and Λ ($\sim 10^{26}$ m). A critical reader is right to ask: “What scale does the torus have, then?”

The answer. T^4 is not an object with a fixed length scale but a *structural property* of the carrier geometry, whose closure invariant ξ is *scale-invariant*. On every length scale the same closure principle manifests with scale-specific resolution. It does not follow that there are “many tori” – it follows that the statement “torus” is, in FFGFT, a *structural* and not a *metric* claim. What is the same on every scale is the geometric closure; what differs from scale to scale is the concrete resolution in which this closure becomes visible.

Empirical trace. This scale-structural reading is not mere stipulation. It is supported by the appearance of ξ as *the same dimensionless constant* in entirely different length contexts:

- lepton-mass cascade (ξ in sub-femtometre physics),
- Higgs formula $\xi = \lambda_h^2 v^2 / (16\pi^3 m_h^2)$ (electroweak sector),
- fine-structure constant $\alpha = e^2 \mu_0 c / (4\pi \hbar)$ (atomic physics; that $\alpha = \xi(E_0/1 \text{ MeV})^2$ reproduces the measured value is a property of the anchor E_0 , not an independent confirmation, Doc. 261),
- LiNbO₃ dispersion difference $\Delta n = \sqrt{3} \xi^{1/2}$ (material physics at μm scale; Doc. 167: $\sqrt{3} \xi^{1/2} = \sqrt{4 \times 10^{-4}} = 0.02$ exactly, measured $2.156 - 2.136 = 0.020$ with 2–3 significant digits),
- Koide relation $\xi_{\text{Koide}} \approx 1.372 \times 10^{-4}$ (lepton ratios).

That the same number appears in five mutually independent physical contexts at very different length scales is empirically non-trivial and is exactly what a scale-invariant closure structure would lead one to expect.

Structural parallel to the renormalisation group. Standard physics already knows this pattern: renormalisation-group methods, asymptotic freedom of QCD, effective field theories work with the same idea – that the same theoretical structure is effective at different scales, with scale-dependent effective manifestations. FFGFT is not outside this tradition here; it makes the pattern explicit: ξ is the scale-invariant quantity, the length scale is the system-dependent resolution. This is also consistent with Doc. 182, in which it is set out that FFGFT recognises *no universal maximum extent* – each scale has a system-specific upper bound.

The methodological point. This observation connects to the reading of $\tilde{T}$ and energy as reciprocal readings developed in § 2: just as $\tilde{T}$ and m are not two different quantities but two scaling axes of the same action, the tori on different length scales are not different structures but the same structural property on different resolution axes. The universality of ξ is the operational manifestation of this identity.

.11 Fractality as an inside-view phenomenon

FFGFT contains a fractal correction structure (factor K_{frak} , appearing in π -gates and in the projection to absolute SI values, Doc. 261). Until now this fractality has been treated in the corpus as a geometric property of T^4 . The discussion of time-as-mode shows that a more precise reading is possible:

As long as time is a discrete mode on the torus, there is no genuine self-iteration. A fractal recursion requires the application of an operation to its own result – and that requires a direction in which “before” and “after” can be distinguished. On a closed torus mode this direction does not exist. It arises only when the mode is decompactified into a directed, quasi-continuous time.

In this reading, the fractal structure is not an ontological feature of the underlying geometry but a *phenomenon of the decompactified time description*. It appears as soon as one enters the mode “directed time” – that is, as soon as one adopts an inside view in which “before” and “after” can be distinguished.

Structural parallel to the Born rule. This position is not isolated. It is structurally parallel to the Born-rule reading from Doc. 230 (§ Operational equivalence): there too, the apparent randomness of quantum measurement arises only in the transition from the deterministic

T^4 dynamics to decompactified time. Both phenomena – apparent randomness and fractal inside-geometry – are consequences of the *same decompactification*, not independent findings.

This structural parallel is a non-trivial result: two phenomena that stand unconnected in standard physics (quantum randomness and fractal corrections) appear in FFGFT as two manifestations of a single cause.

The computational proof. The reading above is not only an ontological clarification; it is a *computationally verifiable result*. In natural units ($\hbar = c = 1$, dimensionless ratios), the entire FFGFT description is *correction-free* – no fractal corrections occur, neither K_{frak} nor any other adjustments. Only in the translation to SI units – the moment in which the energetic action is projected into dimensional quantities with separated units (second, joule, metre) – do the fractal corrections appear. This is set out in detail in Doc. 261 (§ Static geometry and SI projection): as long as one computes in natural units and pure ratios, the description is static and correction-free (Layer 1, compact T^4 , frozen recursion). c , $\hbar$, ε_0 become real only upon decompactification of the intrinsic time winding $\tilde{T}$ into arrow time t (Layer 2, embedding cost).

This shifts the epistemic status of the fractal correction: it is *not an additional term in the theory* that would have to be added phenomenologically, but an *artefact of the SI projection* that arises precisely in the step in which the compact T^4 description is translated into the decompactified SI reading. Whoever computes in natural units stays in the Layer-1 description – and sees no corrections. Whoever computes in SI necessarily moves to the Layer-2 description – and sees the corrections, because they describe exactly what the projection forces in terms of action.

This is the computational proof that fractality is a translation phenomenon and not an ontological property of the carrier. The test is simple and already carried out within the theory itself: the same physical statement formulated in two unit systems. One shows the corrections; the other does not. The distinguishing element is not the physics but the *descriptive choice*.

.12 Inside-view metaphor: brain folds and light

Preliminary note on admissibility. The following metaphor is introduced here *explicitly*, because an openly named metaphor remains controllable, while a hidden allusion invites suspicion of esotericism. It is an *inside-view metaphor*, not an ontological claim about T^4 . It does not say " T^4 looks like a brain"; it says something more specific and weaker.

The metaphor. Once time is decompactified, an observer inside the geometry experiences not the simple structure of the carrier but a fractally folded experiential geometry – with folds, fold-curvatures, and self-similarities on several scales. The folded structure of a brain – many-times wound, self-similar, geometrically rich on a small volume – is a familiar image for exactly this kind of inside structure. It is not *what* T^4 is, but it is an appropriate visualisation of what the *inside view of decompactified time* sees.

The light example sharpens the point. Viewed from outside – in the undistorted description – light moves in a straight line. Within the deformed geometry, however, the path actually traversed is different: it follows the folds and windings, and for the observer situated inside, *this* winding path is the real one, not the idealised straight line of the outside description. The standard reading – straight light paths in Euclidean space – is, in this view, the *outside description*, which holds in the limit of undistorted geometry; the inside view is the description in which we actually live and think.

The clean framing – what the metaphor delivers and what not.

- *What it delivers*: It illustrates that a fractally complex inside-experiential world is compatible with a simple outside carrier structure. It makes plausible why our phenomenal world *appears* complex while its carrier is simple.
- *What it is not*: It is not a claim that the universe is “like a brain”, or that consciousness is “fractal cosmic structure”, or anything similar. Such statements would overstep the frame from metaphor to ontology and would not be defensible within FFGFT.
- *Where it applies*: Only in the transition “discrete mode $\rightarrow$ decompactified time”. Outside this transition – in the pure T^4 description – there is no fractal inside-geometry to describe. The metaphor applies to the inside view, not to T^4 itself.

Within this delimitation the metaphor is methodologically admissible and didactically valuable. It closes a gap that Doc. 162 left open: Doc. 162 showed what *no* form of visualisation can deliver (spatial intuition); this document shows what a *deliberately delimited* inside-view metaphor can deliver (rendering plausible the relationship between a simple outside structure and a complex inside world).

.13 Consequences

From the four preceding sections, three statements follow that individually already stand in the corpus, but are brought together here for the first time as a common structural line:

1. **Time is not distinguished.** It is a mode among four equally periodic coordinates on T^4 . Phenomenal time-experience is the inside view of decompactified time.
2. **Mass is deformation of the carrier.** What GR describes as spacetime curvature is, on T^4 , a local deformation of the closed geometry – with standard GR as the linear, locally inertial approximation.
3. **Fractality is an inside-view phenomenon.** The fractal structure that FFGFT sees appearing in K_{frak} , in the lepton-mass cascade, and in the π -gates is a phenomenon of decompactified time, not a property of the carrier itself. Parallel to the Born-rule reading from Doc. 230, two otherwise unconnected phenomena – quantum randomness and fractal corrections – thus appear as two manifestations of the same cause.

Methodological position. This document tightens no empirical prediction and changes no calculation. It is a *reading clarification*: the structural relationship between the T^4 carrier, decompactified time, and phenomenal experiential world is made explicit. The practical use is that the metaphor of brain folds – and similar inside-view images – now have a cleanly framed standing in the corpus and need no longer operate as unexplained allusions.

.14 Summary

On T^4 , time is not a stage but a mode. Mass is not an object on a spacetime but a deformation of the carrier. Fractality is not a property of the carrier but a phenomenon arising in the transition to decompactified time – structurally parallel to the Born rule and the apparent randomness of quantum measurement. Within this transition the brain-folds metaphor is an admissible inside-view visualisation – not as a claim about T^4 , but as an image for how a simple outside structure becomes a fractally wound inside world.

The outside description is simple. The inside world in which we live is wound. The two descriptions are not in contradiction – they are two sides of the same transition.